VENU (15 s). Good afternoon. We built a fuzzer, a dynamic-analysis tool that writes test queries for PostgreSQL and decides by itself when an answer is wrong. Then we generalized it to four more SQL engines. Eleven PostgreSQL bugs reported, nine fixed upstream, six with our patches; three more fixes filed to DuckDB, Turso and DataFusion.

VENU (30 s). PostgreSQL is one of the most used databases and its planner changes every week. The worst bugs return wrong rows silently. A fuzzer can run millions of queries; the hard part, the oracle problem, is to know when an answer is wrong. We decide by comparison: equivalent questions must give equal answers.

GOWTHAM (60 s). Five steps. One: derive a test case from our typed grammar; types make every query valid, so the server spends its time planning, not rejecting syntax. Two: rewrite the tree into checks; for example TLP splits a query into the rows where a condition is true, false and NULL, and the parts must add up to the whole. Three: run on master, under other planner settings and on release 18.6. Four: triage shrinks the failure on the tree and finds which setting or rewrite flips it. Five: an LLM reads recent commits and proposes grammar, but PostgreSQL decides what enters.

BHUVAN (30 s). Read the middle column: every oracle found something. Two bugs came only from comparing with the last release, one only from the sub-select rewrite, one only from the hang oracle, one only from the estimate oracle. Nine are fixed upstream; six of the fixes are our patches.

VENU (20 s) JSON_TABLE: it compared rows with zero columns; master crashed, releases answered wrong; our patch is back-patched to 17. BHUVAN (20 s) NOT IN: a new optimization assumed values cannot be NULL, but a cast can create one; the release disagreed, so the cross-version oracle caught it; our patch is committed. GOWTHAM (20 s) parallel Append: found through the LLM arm; the planner kept a parallel plan that was no longer safe; the assertion in the server log was the signal; fixed and back-patched to 14.

VENU (45 s). We made the tool target-independent. Each engine gets a driver and a dialect overlay on the same grammar; a miner drafts the overlay from failure rates. Instead of PostgreSQL's planner settings we observe which optimizer rules fire, which works on any release build. Z3 makes metamorphic checks non-vacuous, learned normalizers let us compare two engines, and the hunt loop goes from campaign to a filing package on its own.

GOWTHAM (20 s) Turso: only the comparison with SQLite showed the wrong value; the aggregate read its fraction from the wrong register. BHUVAN (20 s) DuckDB: the internal-error oracle; a 2.0.0 change made the binder copy an expression that cannot be copied. VENU (20 s) DataFusion: an optimizer rule leaked a plan-only column into UPDATE and DELETE. All three pull requests carry a regression test and pass the engine's full test suite.

BHUVAN (30 s). Typed generation reaches far more backend code than random mutation, and the oracles find bugs a crash fuzzer cannot see. Our main lesson: precision of the oracles is the limit. Thank you.